Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

Methods of Engineering Work

Academic Year 2026/2027

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Application for Applied Research (Call VV-2024)

Item VV-A1-01: Project ID

APVV-24-0000

Item VV-A1-02: Date of submission

1.11.2026

Item VV-A1-03: Project Title

Implementation of Post-Quantum Cryptography in V2X Systems

Item VV-A1-04: Project Acronym

V2X-PQC

Item VV-A1-05: R&D specialization

Informatics (Cybersecurity)

Item VV-A1-06: R&D characterization

Applied research

Item VV-A1-07: Project start

1.1.2027

Item VV-A1-01: Project end

31.12.2028

Item VV-A1-09: Project Annotation

The advent of quantum computers poses an imminent security threat to cooperative intelligent transport systems, making a transition to post-quantum cryptography (PQC) unavoidable. Standardized PQC signature schemes fit vehicle-to-everything (V2X) communication poorly. ML-DSA (Dilithium) produces 2,420-byte signatures that can overload communication channels in dense traffic, while Falcon, with 666-byte signatures, saves bandwidth but its computationally demanding verification threatens the critical 10 ms latency limit at busy intersections. Existing evaluations rarely reflect realistic traffic, so it remains unclear which PQC scheme, if any, satisfies V2X constraints.

The main objective of the V2X-PQC project is to systematically benchmark PQC signature algorithms under realistic V2X conditions and, depending on the findings, to design a new algorithm fit for this use case. To this end, the team will extend the open-source AWSIM simulator with a V2X communication layer and a message verification module. This will enable reproducible experiments with standardized BSM messages, varying traffic density, message rates of up to 1,000 messages per second, and the limited processing capacity of verifying devices.

The project will deliver a simulation framework, a benchmark of candidate PQC schemes (including Dilithium and Falcon) measuring signature size, channel load, verification latency and throughput, and a research paper summarizing the results. Building on the identified bottlenecks, the project will also propose and evaluate an original PQC algorithm, or an optimization of an existing one, targeting verification latency below 10 ms at 1,000 messages per second and substantially less transmitted cryptographic data than Dilithium. The results will give vendors, operators and standardization bodies evidence-based guidance for adopting quantum-safe V2X security.

VV-C – Budget of the Project in EUR

Applicant

Budget category	2027 (EUR)	2028 (EUR)	Total (EUR)
01. Direct running costs	32,619.20	33,019.20	65,638.40
02. Wage and other personal costs	21,600.00	21,600.00	43,200.00
03. Social and health insurance	7,819.20	7,819.20	15,638.40
04. Travel costs	500.00	1,500.00	2,000.00
05. Material	1,500.00	900.00	2,400.00
06. Services	900.00	900.00	1,800.00
07. Energy, water, communications	300.00	300.00	600.00
08. Popularization costs	0.00	0.00	0.00
09. Indirect costs	1,500.00	1,500.00	3,000.00
10. Total running costs	**34,119.20**	**34,519.20**	**68,638.40**
Total costs from APVV	**34,119.20**	**34,519.20**	**68,638.40**
Financing from other sources	0.00	0.00	0.00
State (outside APVV)	0.00	0.00	0.00
Foreign	0.00	0.00	0.00
Private	0.00	0.00	0.00
Total costs	**34,119.20**	**34,519.20**	**68,638.40**

List of Expected Costs Covered by APVV

No.	Expected costs
02	**Wage and other personal costs – 43,200.00 EUR** Principal Investigator: 500 hours/year at 18 EUR/hour (18,000 EUR in total); Cryptography Researcher: 300 hours/year at 16 EUR/hour (9,600 EUR); Software/V2X Developer: 300 hours/year at 14 EUR/hour (8,400 EUR); Testing and Evaluation Researcher: 300 hours/year at 12 EUR/hour (7,200 EUR).
03	**Social and health insurance – 15,638.40 EUR** Employer contributions estimated at 36.2% of gross wages: 43,200.00 EUR $\times$ 0.362 = 15,638.40 EUR.
04	**Travel costs – 2,000.00 EUR** 2027: domestic research meetings and technical workshop (500 EUR). 2028: one scientific conference or specialist workshop, travel and accommodation (1,500 EUR).
05	**Material – 2,400.00 EUR** 2027: development peripherals, network interfaces, cables and laboratory consumables (1,500 EUR). 2028: adapters, replacement components and additional experimental materials (900 EUR).
06	**Services – 1,800.00 EUR** 2027: computing services, hosting and technical support (900 EUR). 2028: computing resources, storage and technical services (900 EUR).
07	**Energy, water, communications – 600.00 EUR** Electricity and network communications related to testing and computations: 300 EUR per year.
08	**Popularization costs – 0.00 EUR** No funds requested.
09	**Indirect costs – 3,000.00 EUR**

No.	Expected costs
	Administrative, accounting and shared laboratory support: 1,500 EUR per year.

Summary Budget of the Project

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VV-D – Project Schedule

Begin of phase	End of phase	Phase name
01.01.2027	30.04.2027	Analysis of V2X security requirements and post-quantum signature algorithms
01.05.2027	31.10.2027	System architecture design and development of the initial PQC software prototype
01.11.2027	30.04.2028	Integration of cryptographic mechanisms into the simulated V2X environment
01.05.2028	30.09.2028	Performance testing and optimization of signature verification and communication overhead
01.10.2028	31.12.2028	Final validation, evaluation of results and technical documentation

List of Devices in Project

Name	Description
Research workstations	Computers for software development, V2X simulation and cryptographic benchmarking.
Network testing components	Network interfaces, adapters and cables for experimental communication tests.
Embedded development modules	Small-scale development boards for performance measurements of cryptographic implementations.
Computing infrastructure	Computing and storage resources for simulations, processing test data and evaluating results.